# Day 13: Pressure, Density, Buoyancy

SHS Physics Club Summer Camp 2026

# Overview

**01 HW/Quiz**

Brief MCQ review quiz, go over homework questions

**02 Density**

Density for solids and liquids

**03 Pressure**

Pressure, air pressure, water pressure

**04 Buoyancy**

Buoyancy force, floating/sinking, advanced buoyancy problems

# 1 - HW/Quiz

# HW/Quiz

Homework Questions?

# HW/Quiz

- **Quiz**
  - 10 min
  - 5 MCQ

# 2 - Density

# Density

- **How much mass is packed into a certain volume**
  - **Mass to volume ratio**
  - **ρ = m / V**
  - **What is heavier, a gram of metal or a gram of plast**
- **Specific gravity:**
  - **Nothing to do with gravity**
  - **Ratio of a substance’s destiny to the density of water**
- **Average density**
  - **An object's density might not be uniform**
  - **Total mass over total volume!**

Fresh water: 1000 kg/m³
Seawater: about 1025 kg/m³
Ice: about 917 kg/m³
Aluminum: about 2700 kg/m³
Iron: about 7870 kg/m³
Air: about 1.2 kg/m³

# Practice Problems

A metal block has a mass of 18 kg and a volume of 0.006 $m^3$.

What is the density of the block?

A hollow plastic ball has a total mass of 2.4 kg and a volume of 0.0040 $m^3$. The plastic itself has a density greater than water, but the ball contains air.

Find the average density of the entire ball.

# 3 - Pressure

# Pressure

- **Force over area**
  - **P = F/A (pascal)**
  - **Needle vs bed of nails**
- **Water pressure**
  - **Water has a weight, so whatever is underneath experiences some form of pressure**
  - **Same thing as air pressure, but air is less dense**
  - **Derive pressure in terms of height:**

- **Absolute pressure**
  - **Gauge pressure + atmospheric pressure (I'll explain)**

# Practice

A person stands on the floor and then lifts one foot. What happens to the pressure exerted on the floor?

Three containers have different shapes. Each contains water filled to the same height. Points A, B, and C are located at the bottom of each container.

- Container A is narrow.
- Container B is wide.
- Container C is wider at the top and narrow at the bottom.

Compare the water pressure at points A, B, and C.

# Practice

A rectangular aquarium is filled with fresh water to a depth of 1.5 m. A viewing window on one side is 0.80 m wide and 0.60 m tall. The top of the window is 0.40 m below the water’s surface.

Find:

a. The depth of the center of the window
b. The water pressure at the center of the window
c. The average force exerted by the water on the window

# 4 - Buoyancy

# Buoyancy Force

- **Pressure scales with depth, so more pressure from the bottom than from the top**
  - **$F_B$ = weight of water displaced (derive equation)**
- **Floating: when the weight of a submerged object is less than the bouyancy force**
- **Sinking: quite the opposite**
- **Neutral buoyancy: when both are =**
- **Nautilus?**



# Practice

A block with a volume of 0.015 $m^3$ is completely submerged in fresh water.

a. What volume of water does the block displace?
b. What is the buoyant force acting on the block?

You slowly dip one finger into a cup of water without touching the bottom or sides.

What happens to the reading on a scale underneath the cup?

A. It decreases
B. It increases
C. It stays the same
D. It becomes zero

An ice cube floats in a full glass of water. After the ice completely melts, what happens to the water level?

A. It rises
B. It falls
C. It stays the same
D. The glass overflows

# Floating - explanation

# Practice

A metal block has a mass of 12 kg and a volume of 0.0030 $m^3$. It hangs from a spring scale and is completely submerged in fresh water without touching the bottom.

a. Find the block’s weight.
b. Find the buoyant force.
c. Find the tension in the rope.
d. What does the spring scale read?

A wooden block has a density of 720 $kg/m^3$ and floats in fresh water.

a. What fraction of the block is submerged?
b. What percentage of the block is above the water?

# See you next class!